

Real- z Check of the Shortcut Sign in Eq. (29)

Comparison with the direct stochastic shortcut matrix

Short answer. The quick check locates the sign issue already in Eq. (29). Used literally, the printed Eq. (29) reproduces the original Eq. (32), but both disagree with the direct stochastic shortcut matrix. With the stochastic shortcut sign, the Eq. (29) ratio and the corrected Eq. (32) agree with the direct matrix to roundoff.

Direct Matrix Benchmark

The benchmark is the finite-state transition matrix for the stochastic shortcut model itself, not Eq. (29) or Eq. (32). Start from the lazy ring matrix

$$T_0(i, i) = 1 - q, \quad T_0(i, i \pm 1) = q/2,$$

and move the shortcut mass

$$\lambda = \beta(1 - q)$$

from the $u \rightarrow u$ self-loop to the directed edge $u \rightarrow v$:

$$\tilde{T} = T_0 + \lambda(E_{uv} - E_{uu}).$$

Then compute the resolvent and the first-passage ratio directly:

$$\tilde{S}(z) = (I - z\tilde{T})^{-1}, \quad \tilde{F}_{n_0}(v, z) = \frac{\tilde{S}_{n_0 v}(z)}{\tilde{S}_{vv}(z)}.$$

Sign Implied by the Matrix

The direct stochastic matrix gives

$$I - z\tilde{T} = (I - zT_0) + z\lambda(E_{uu} - E_{uv}).$$

Writing $P(z) = (I - zT_0)^{-1}$, the corresponding rank-one update has the sign pattern

$$\tilde{P}_{ij}(z) = P_{ij}(z) - \frac{z\lambda P_{iu}(z)\{P_{uj}(z) - P_{vj}(z)\}}{1 + z\lambda\{P_{uu}(z) - P_{vu}(z)\}}.$$

Thus the stochastic shortcut $u \rightarrow v$ gives a plus sign in this denominator and the corresponding minus sign in the correction term. The printed Eq. (29) has the opposite pattern: a plus sign in the correction term and a $1 - z\lambda\{P_{uu}(z) - P_{vu}(z)\}$ denominator. That is the inverse for the opposite rank-one defect, not for removing probability mass from $u \rightarrow u$ and adding it to $u \rightarrow v$. This is why the sign issue is already in Eq. (29).

Numerical Check

The grid uses the figure parameters $N = 10$, $q = 2/3$, $\beta = 2/7$, $u = 5$, $v = 0$, plus a second symmetric case, with several starts n_0 and real values of z . Site labels here are zero-based.

Main sign check

Quantity compared over the grid	Maximum absolute error
Literal printed Eq. (29) ratio vs direct stochastic matrix	3.669746×10^{-1}
Original Eq. (32) vs direct stochastic matrix	3.669746×10^{-1}
Stochastic-sign Eq. (29) ratio vs direct stochastic matrix	2.220446×10^{-16}
Corrected Eq. (32) vs direct stochastic matrix	2.775558×10^{-16}
Literal printed Eq. (29) ratio vs original Eq. (32)	1.318390×10^{-16}

Control checks

Quantity compared over the grid	Maximum absolute error
Eq. (19) $W_{n_0}(u, z)$ vs absorbing matrix	1.043610×10^{-14}
Eq. (28) $W_u(u, z)$ vs absorbing matrix	1.687539×10^{-14}

The control checks show that the discrepancy is not coming from the $W_{n_0}(u, z)$ or $W_u(u, z)$ inputs used in Eq. (32).

Representative Values

n_0	z	direct matrix	printed Eq. (29) / original Eq. (32)	stochastic sign / corrected Eq. (32)
2	0.80	0.197025316799	0.173752283343	0.197025316799
4	0.80	0.095953324694	-0.033502923905	0.095953324694

Interpretation. For $0 < z < 1$, a first-passage generating function has non-negative coefficients, so it should not become negative. The negative value in the second row is a useful sanity check that the printed sign is not the stochastic shortcut sign for this $u \rightarrow v$ model.

Conclusion

The printed Eq. (29) and the original Eq. (32) are internally consistent with each other. The problem is that this shared sign convention differs from the direct probability-conserving stochastic shortcut matrix. Therefore the sign issue is already present in Eq. (29) under the stochastic $u \rightarrow v$ interpretation.